

Matrix & Vector Multiplication :

* Matrix and Vector Multiplication

→ A Linear Combination is a weighted sum of vectors such that it results in a new vector within same space.

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n = \alpha_1 \begin{bmatrix} a_{11} \\ a_{12} \\ a_{13} \\ \vdots \\ a_{1n} \end{bmatrix} \in \mathbb{R}^n + \alpha_2 \begin{bmatrix} a_{21} \\ a_{22} \\ a_{23} \\ \vdots \\ a_{2n} \end{bmatrix} \in \mathbb{R}^n + \alpha_3 \begin{bmatrix} a_{31} \\ a_{32} \\ a_{33} \\ \vdots \\ a_{3n} \end{bmatrix} \in \mathbb{R}^n + \dots + \alpha_k \begin{bmatrix} a_{k1} \\ a_{k2} \\ a_{k3} \\ \vdots \\ a_{kn} \end{bmatrix} \in \mathbb{R}^n$$

→ Above Concept can be extended to a Matrix & vector multiplication

Example: let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Compute Ax

→ Approach 1 (dot product)

$$Ax = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \times 3 + 0 \times 4 \\ 0 \times 3 + 1 \times 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Approach 2 (Linear Combination of Column Vectors) ← Geometrically more intuitive

$$Ax = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Example: let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ and $x = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. Compute Ax

→ Approach 1 (dot product)

$$Ax = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 0 + 3 \cdot 1 \\ 4 \cdot 1 + 5 \cdot 0 + 6 \cdot 1 \\ 7 \cdot 1 + 8 \cdot 0 + 9 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \\ 16 \end{bmatrix}$$

Approach 2 (Linear Combination of Column Vectors)

$$Ax = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} + 1 \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \\ 16 \end{bmatrix}$$

Example: let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ and $x = \begin{bmatrix} 6 \\ 5 \\ 4 \end{bmatrix}$. Compute Ax

→ Approach 1 (dot product)

$$Ax = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 6 \\ 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \cdot 1 + 5 \cdot 2 + 4 \cdot 3 \\ 4 \cdot 6 + 5 \cdot 5 + 6 \cdot 4 \\ 7 \cdot 6 + 8 \cdot 5 + 9 \cdot 4 \end{bmatrix} = \begin{bmatrix} 28 \\ 73 \\ 118 \end{bmatrix}$$

Approach 2 (Linear Combination of Column Vectors)

$$\begin{aligned} Ax &= \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 6 \\ 5 \\ 4 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} + 5 \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} + 4 \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} \\ &= \begin{bmatrix} 6 \\ 24 \\ 42 \end{bmatrix} + \begin{bmatrix} 10 \\ 25 \\ 40 \end{bmatrix} + \begin{bmatrix} 12 \\ 24 \\ 36 \end{bmatrix} \\ &= \begin{bmatrix} 28 \\ 73 \\ 118 \end{bmatrix} \end{aligned}$$

Example: $\begin{bmatrix} 11 & 12 & 13 \\ 14 & 15 & 16 \\ 17 & 18 & 19 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 24 \\ 30 \\ 36 \end{bmatrix}$ Compute a, b and c .
← $Ax = b$

→ Linear combination of column vectors approach -

$$a \begin{bmatrix} 11 \\ 14 \\ 17 \end{bmatrix} + b \begin{bmatrix} 12 \\ 15 \\ 18 \end{bmatrix} + c \begin{bmatrix} 13 \\ 16 \\ 19 \end{bmatrix} = \begin{bmatrix} 24 \\ 30 \\ 36 \end{bmatrix}$$

Trying some trial and error for (a, b, c) , we get

$$\boxed{a=1 \quad b=0 \quad c=1}$$

* Multiplication of 2 Matrices

Example: Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Compute AB

→ Approach 1 (dot product)

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 \times 0 + 2 \times 1 & 1 \times 1 + 2 \times 0 \\ 3 \times 0 + 4 \times 1 & 3 \times 1 + 4 \times 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

Example: Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$. Compute AB

→ Approach 1 (dot product)

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 \times 2 + 2 \times 1 & 1 \times 1 + 2 \times 0 \\ 3 \times 2 + 4 \times 1 & 3 \times 1 + 4 \times 0 \\ 5 \times 2 + 6 \times 1 & 5 \times 1 + 6 \times 0 \end{bmatrix}$$
$$= \begin{bmatrix} 4 & 1 \\ 10 & 3 \\ 16 & 5 \end{bmatrix}$$

→ Approach 2 (Linear Combination of column vectors)

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$

① we can write $B = [b_1 \ b_2 \ b_3 \ \dots \ b_p]$

$$\begin{aligned} \text{② } AB &= A [b_1 \ b_2 \ b_3 \ \dots \ b_p] \\ &= [Ab_1 \ Ab_2 \ Ab_3 \ \dots \ Ab_p] \end{aligned}$$

$$AB = [Ab_1 \ Ab_2 \ Ab_3 \ \dots \ Ab_p]$$

↑ where $b_1, b_2, b_3 \dots b_p \in \mathbb{R}^n$

Example: Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Compute AB using approach 2.

$$\begin{aligned} \rightarrow AB &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 4 \end{bmatrix} & 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ 4 \end{bmatrix} \end{bmatrix} \\ &= \begin{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} & \begin{bmatrix} 1 \\ 3 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \end{aligned}$$

Example: Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$. Compute AB using approach 2.

$$\rightarrow AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} & 1 \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \end{bmatrix}$$

$$\begin{aligned} &= \begin{bmatrix} \begin{bmatrix} 2 \\ 6 \\ 10 \end{bmatrix} + \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} & \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \end{bmatrix} \\ &= \begin{bmatrix} 4 & 1 \\ 10 & 3 \\ 16 & 5 \end{bmatrix} \end{aligned}$$